

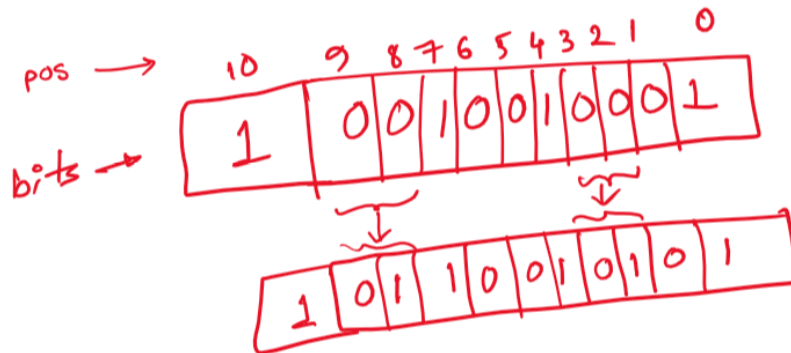
Online Assignment on Bitwise operations (A1)

Solve the problem below. You have **20 minutes**. After you are done, rename the file containing your source code as your student ID (so, if your student ID is 2005001, the name of your file should be **2005001.c**). Then submit that file to Moodle. Make sure you submit a file containing the source code.

Failure to not follow these instructions will result in penalties.

Problem 1.

- Write a C program that takes a positive integer as input and replaces all occurrences of **two consecutive 0's**, where the **first 0 is in odd position**, with **01**. Print the bits of the given integer and the integer after replacement.



Input	Output
24 (011000)	11000 11001
4 (0100)	100 101
43 (101011)	101011 101011

N.B.

- ★ You **must** use **bitwise operators** to solve the problem.
- ★ You **can** assume that the provided input will **always** be valid.
- ★ **Build the decimal number for your output first** and then print the bits of the number.